

DW_pricod

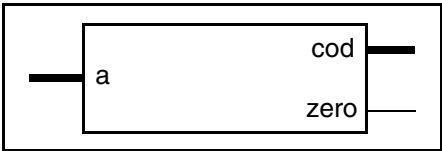
Priority Coder

Version, STAR, and myDesignWare Subscriptions: [IP Directory](#)

Features and Benefits

- Parameterized word length
- Inferable using a function call

Revision History



Description

The `cod` output of `DW_pricod` is a coded one-hot value of the `a` input vector with a 1 at the most significant (left-most) non-zero bit position of `a`. All lower order bits (to the right) from the first occurrence of a 1 on the `a` input port are “don’t care.” The `zero` output indicates whether all bits of input `a` are 0. If a 1 is not found and only 0s are present, the resulting value of `cod` is all 0s and the value of `zero` is 1.

Table 1-1 Pin Description

Pin Name	Width	Direction	Function
a	<i>a_width</i>	Input	Input vector
cod	<i>a_width</i>	Output	One-hot coded value of a.
zero	1	Output	All-zero flag (= 1 if all bits of a are 0)

Table 1-2 Parameter Description

Parameter	Values	Description
a_width	≥ 1	Vector width of input a

Table 1-3 Synthesis Implementations

Implementation Name	Function	License Feature Required
rtl	Synthesis model	DesignWare
cla	Synthesis model	DesignWare

Table 1-4 Simulation Models

Model	Function
DW01.DW_PRICOD_CFG_SIM	Design unit name for VHDL simulation
dw/dw01/src/DW_pricod_sim.vhd	VHDL simulation model source code
dw/sim_ver/DW_pricod.v	Verilog simulation model source code

Table 1-5 Truth Table (a_width = 8, cod width = 8)

a(7:0)								cod(7:0)								zero
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1
0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1	0
0	0	0	0	0	0	1	X	0	0	0	0	0	0	1	0	0
0	0	0	0	0	1	X	X	0	0	0	0	0	1	0	0	0
0	0	0	0	1	X	X	X	0	0	0	0	1	0	0	0	0
0	0	0	1	X	X	X	X	0	0	0	1	0	0	0	0	0
0	0	1	X	X	X	X	X	0	0	1	0	0	0	0	0	0
0	1	X	X	X	X	X	X	0	1	0	0	0	0	0	0	0
1	X	X	X	X	X	X	X	1	0	0	0	0	0	0	0	0

Related Topics

- [Math – Arithmetic Overview](#)
- [DesignWare Building Block IP User Guide](#)

HDL Usage Through Function Inferencing - VHDL

```
library IEEE,DWARE;
use IEEE.std_logic_1164.all;
use DWARE.DW_Foundation.all;
-- If using numeric types from std_logic_arith package,
-- comment the preceding line and uncomment the following line:
-- use DWARE.DW_Foundation_arith.all;

entity DW_pricod_func is

    generic (
        func_a_width : POSITIVE := 8);

    port (
        func_a      : in  std_logic_vector(func_a_width-1 downto 0);
        cod_func     : out std_logic_vector(func_a_width-1 downto 0));

end DW_pricod_func;

architecture func of DW_pricod_func is

begin

    -- Function inference of DW_pricod
    cod_func <= DWF_pricod (func_a);

end func;

-- pragma translate_off
configuration DW_pricod_func_cfg_func of DW_pricod_func is
    for func
        end for;
end DW_pricod_func_cfg_func;
-- pragma translate_on
```

HDL Usage Through Function Inferencing - Verilog

```
module DW_pricod_func (func_a, cod_func);

    parameter func_a_width = 8;

    // Passes the width to DW_pricod_function
    parameter a_width = func_a_width;

    `include "DW_pricod_function.inc"

    input  [func_a_width-1 : 0] func_a;
    output [func_a_width-1 : 0] cod_func;

    // Function inference of DW_pricod
    assign cod_func = DWF_pricod (func_a);

endmodule
```

HDL Usage Through Component Instantiation - VHDL

```
library IEEE,DWARE;
use IEEE.std_logic_1164.all;
use DWARE.DW_Foundation_comp.all;
-- If using numeric types from std_logic_arith package,
-- comment the preceding line and uncomment the following line:
-- use DWARE.DW_Foundation_comp_arith.all;

entity DW_pricod_inst is

    generic (
        inst_a_width : POSITIVE := 8);

    port (
        inst_a      : in  std_logic_vector(inst_a_width-1 downto 0);
        cod_inst     : out std_logic_vector(inst_a_width-1 downto 0);
        zero_inst    : out std_logic);

end DW_pricod_inst;

architecture inst of DW_pricod_inst is

begin

    -- Instance of DW_pricod
    U1 : DW_pricod
        generic map (a_width => inst_a_width)
        port map (a => inst_a, cod => cod_inst, zero => zero_inst);

end inst;

-- pragma translate_off
configuration DW_pricod_inst_cfg_inst of DW_pricod_inst is
    for inst
    end for;
end DW_pricod_inst_cfg_inst;
-- pragma translate_on
```

HDL Usage Through Component Instantiation - Verilog

```
module DW_pricod_inst (inst_a, cod_inst, zero_inst);

    parameter inst_a_width = 8;

    input  [inst_a_width-1 : 0] inst_a;
    output [inst_a_width-1 : 0] cod_inst;
    output zero_inst;

    // Instance of DW_pricod
    DW_pricod #(inst_a_width)
        U1 ( .a(inst_a), .cod(cod_inst), .zero(zero_inst) );

endmodule
```

Revision History

For notes about this release, see the [DesignWare Building Block IP Release Notes](#).

For lists of both known and fixed issues for this component, refer to the [STAR report](#).

Date	Release	Updates
September 2024	DWBB_202409.0	■ Updated date and version; no technical changes
December 2023	DWBB_202312.0	■ Updated date and version; no technical changes ■ Added this Revision History table and the document links on this page

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